Due: Tuesday March 29th, during class starting at noon

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1. [**2 pts.**] Show that
$$\frac{1}{\gamma^2}+\frac{v^2}{c^2}=1$$

2. [**2 pts.**] Show that
$$\gamma(v)\times(c+v)=c\sqrt{\frac{c+v}{c-v}}$$

3. The binomial expansion
$$(1+x)^\epsilon = 1+\epsilon x+\frac{1}{2!}\epsilon(\epsilon-1)x^2+\frac{1}{3!}\epsilon(\epsilon-1)(\epsilon-2)x^3+......$$
holds for all $|x|<1$. Please find a way to remember this, so that you can use it without looking it up. (Not just for this module, but for all of physics.)

   (a) [**3 pts.**] Expand γ in powers of v/c, including at least up to order $(v/c)^6$. This expansion is useful for small v, when γ is close to 1.

   [**Extra credits.**] If you truncate the series, you get an approximant for $\gamma(v)$. Plot $\gamma(v)$ as a function of v/c, together with the approximants obtained by keeping the first one, two, three or four terms of the series. This should show how successive approximants do a better and better job of approximating our $\gamma(v)$ function.

   (b) [**3 pts.**] Express v as a function of $1/\gamma$. Expand using the binomial expansion, at least up to order $1/\gamma^6$. This expansion is useful for large γ, when v is close to c. Why?

4. A particle of rest mass M, while at rest in the laboratory, decays into a particle of mass m and speed v, and a photon of frequency f moving in opposite direction. Relativistic momentum and energy is conserved in this process.

   (a) [**2 pts.**] Draw the situations before and after the decay process, as seen in the laboratory frame. Remember to indicate the velocity of each particle.

   (b) [**4 pts.**] Write down the equations for momentum conservation and energy conservation (in the laboratory frame).

(c) [**4 pts.**] Use these equations to calculate m as a function of M and v.

(d) [**3 pts.**] Show that the photon frequency is given by

$$hf = \frac{Mvc^2}{c+v}$$

in the lab frame.

(e) [**3 pts.**] Now work in the frame moving at speed v; in this frame the mass m is at rest after the decay. Draw the situations before and after the decay process, as seen in this frame. Do indicate the speeds.

(f) [**4 pts.**] Write down the equations of momentum conservation and energy conservation in this frame.

(g) [**3 pts.**] Solve for m (as a function of M and v) to show that you get the same expression for m as you did by working in the laboratory frame.

(h) [**3 pts.**] Solve for the photon frequency to show that you get a *different* frequency. What is the factor by which the photon frequency is different?

(i) [**3 pts.**] The photon frequency is different, when seen from a different frame! But you should know this as the Doppler effect. Use the result obtained in class for the relativistic Doppler effect to find the expected Doppler-shifted frequency. Does this match what you obtained above from momentum/energy conservation?

5. [**6 pts.**] A subatomic particle lives for 6×10^{-7} seconds, i.e., it decays 6×10^{-7} seconds after it is created, as seen from its own reference frame. If this particle is created at a speed $v = \frac{4}{5}c$ with respect to earth, how far will it travel before it decays, as seen from the earth frame?

   You can use the approximation $c \approx 3 \times 10^8$m/s.

6. A photon of wavelength λ collides with a stationary electron. After the collision, the photon scatters at an angle θ with respect to the incident direction, and has wavelength λ'. The electron moves with momentum p_e after the collision, in a direction making angle ϕ with the incident direction of the photon.

   (a) [**3 pts.**] Draw the situations before and after, clearly showing everything relevant.

   (b) [**6 pts.**] Write down the equations for energy conservation, and two equations for momentum conservation. (Since momentum is a vector, you have one equation for each relevant direction.)

(c) [**6 pts.**] Show that

$$\lambda' = \lambda + \frac{h}{mc}(1 - \cos\theta)$$

Hint: you don't need to introduce velocity. If you need to transform between energy and momentum, use the relation $E^2 = p^2c^2 + m^2c^4$.